COMMERCEOS ENTERPRISE ERP

System Architecture, Functional Reference & User Flow Manual

Complete Operational Guide for Multi-Store Retail, Wholesale (B2B), Warehouse WMS, and General Ledger Accounting

SALES CHANNELS	OPERATIONAL MODULES	FULFILLMENT NODES	LIVE ORDERS ENGINE
4 Storefronts	11 Modules	4 Warehouses	2,846 Orders

Target Audience: Operations Leads, WMS Teams, Finance Controllers, System Administrators
Platform Release: CommerceOS v2.4 (Enterprise Edition)
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NOTE

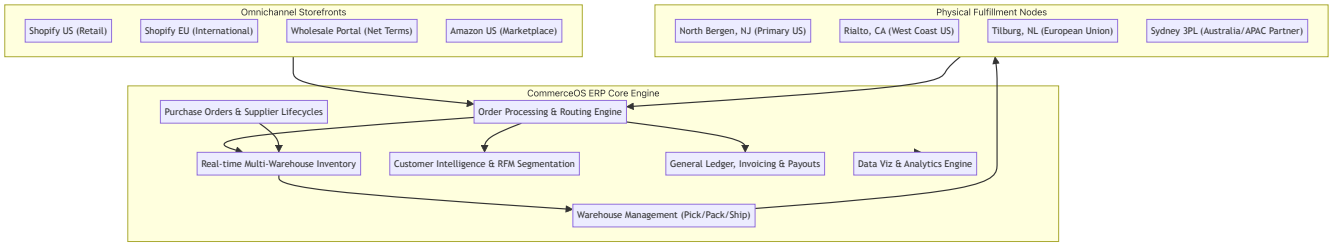
CommerceOS is a high-performance, modular enterprise eCommerce Operating System (ERP) engineered for multi-store retail, wholesale (B2B), and marketplace operations. This manual provides a single source of truth for end users, operations managers, supply chain leads, finance teams, and developers.

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1. Executive Overview & ERP Architecture

CommerceOS replaces fragmented standalone SaaS applications (separate inventory managers, warehouse management systems, CRM databases, accounting spreadsheets, and order desks) with a unified, tightly coupled data engine and cohesive operational UI.



Architectural Tenets

1. **Zero Data Latency:** Inventory reservation, customer order tracking, and financial ledgers update synchronously across all channels.
2. **Deterministic State Transitions:** Orders, RMAs, Transfers, and POs adhere to strict state machines with validation guards against illegal states (e.g. shipping unpicked orders or refunding uninspected goods).
3. **Ergonomic Density:** Built on standard 20px card insets, tabular numeric alignment, dual light/dark modes, and single-key shortcuts for rapid operations desk execution.

2. Multi-Store & Omnichannel Model

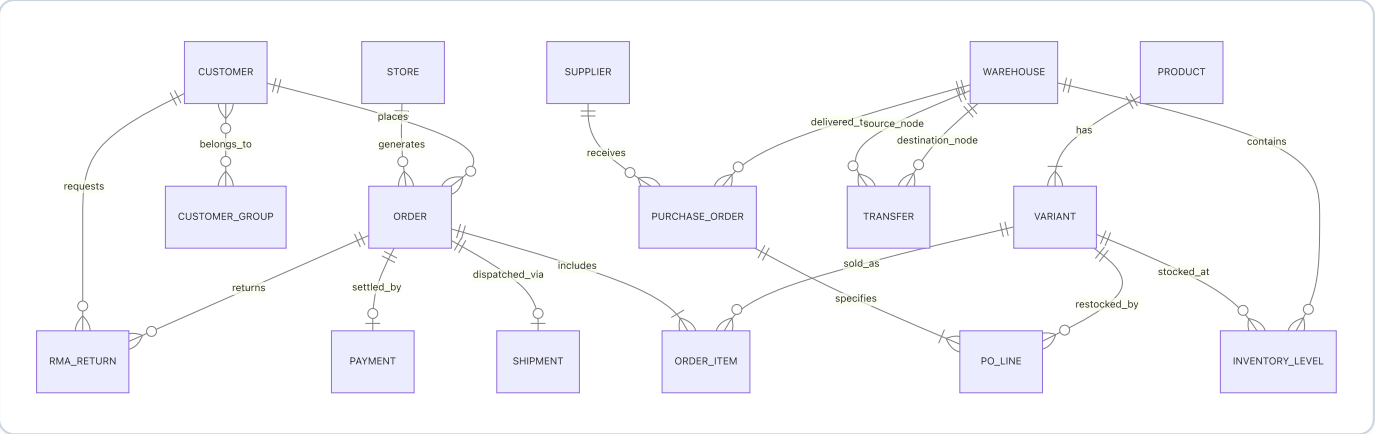
CommerceOS natively partitions and aggregates data across 4 distinct sales channels:

STORE CHANNEL	CHANNEL KEY	BUSINESS MODEL	PRIMARY CURRENCY	FULFILLMENT STRATEGY	REVENUE SHARE
Fernhill US	`us`	Direct-to-Consumer (DTC)	USD ($)	North Bergen (NJ) & Rialto (CA)	~26.0%
Fernhill EU	`eu`	International DTC	EUR (€)	Tilburg Hub (Netherlands)	~19.7%
Wholesale	`wholesale`	B2B Commercial Accounts	USD ($) Net 30/45	Cross-docking & North Bergen bulk	~48.9%
Amazon US	`amazon`	FBM / 3P Marketplace	USD ($)	Rialto (CA) dedicated bin allocation	~5.4%

Users can filter any view across the system to a single store channel using the top navigation bar or analyze consolidated multi-store revenue ($820,443/month).

3. System Data Architecture & Entity Relationship Model

The core relational backbone connects Products, Inventory Allocations, Customer CRM records, Orders, Warehouses, and Financial Transactions:



Data Synchronization Layer

- `catalog.js`: Master catalog of 42 high-volume active SKUs, product options, hierarchical collections, and customer reviews.
- `inventory.js`: Exact stock quantities across all 4 fulfillment centers: On Hand, Allocated, In Transit, and Available.
- `orders.js`: 2,846 live orders with deterministic chronological status, carrier tags, line items, and risk scoring.
- `fulfillment.js`: Queue generators for pick lists, packing station bins, shipping slots, and carrier manifests.
- `procurement.js`: Vendor records, open PO contracts, goods receipt notes (GRN), and Return to Vendor (RTV) claims.
- `finance.js`: Merchant fee deductions, payout settlements, customer invoices, corporate expense allocations, and reconciliation tables.

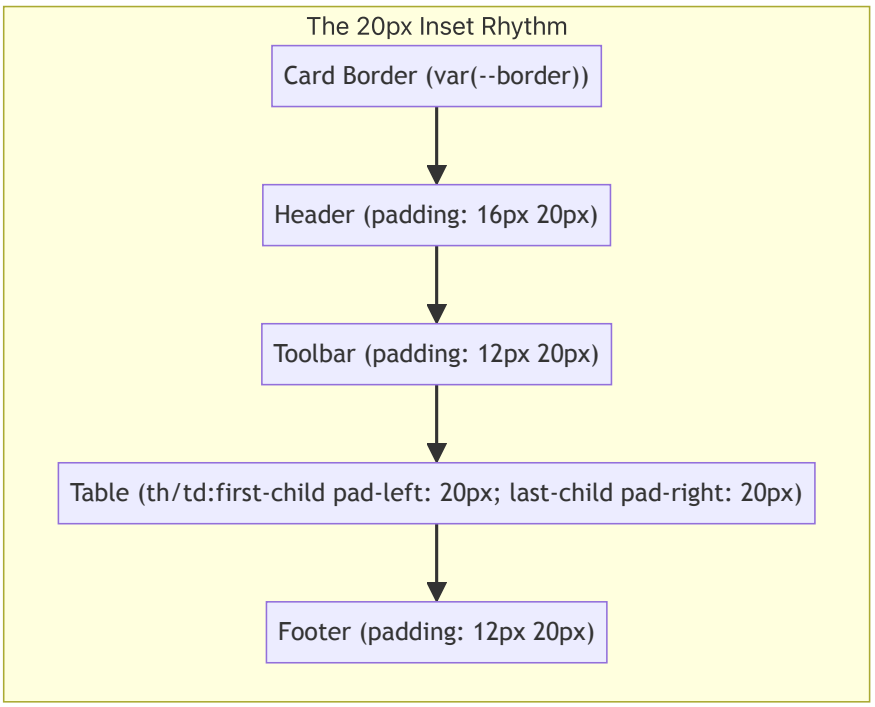
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4. User Interface, Design System & Ergonomics

The CommerceOS interface is governed by a unified Design System that guarantees visual rhythm, responsiveness, and speed.

Grid Rhythm & Spacing Standards

- **Standard Card Inset (`--space-20`):** Every card header (`.cx-card-head`), card body (`.cx-card-body`), table margin, toolbar, and drawer panel starts its content precisely **20px in**.
- **Container Gutters:**
- **Desktop (≥1280px):** 32px top-bar padding, 24px/32px page container padding, max-width 1600px.
- **Tablet (768px–1279px):** 24px top-bar padding, 24px page padding, 64px icon rail sidebar.
- **Mobile (<768px):** 16px top-bar padding, 16px page padding, drawer navigation, scroll-snapped horizontally swipeable KPI tiles.
- **Tabular Figures:** All currency amounts, SKUs, timestamps, and inventory units utilize monospaced numeric formatting (`font-feature-settings: "tnum" 1`) to eliminate visual column jitter.


The 20px Inset Rhythm
Card Border (var(--border))
Header (padding: 16px 20px)
Toolbar (padding: 12px 20px)
Table (th/td:first-child pad-left: 20px; last-child pad-right: 20px)
Footer (padding: 12px 20px)

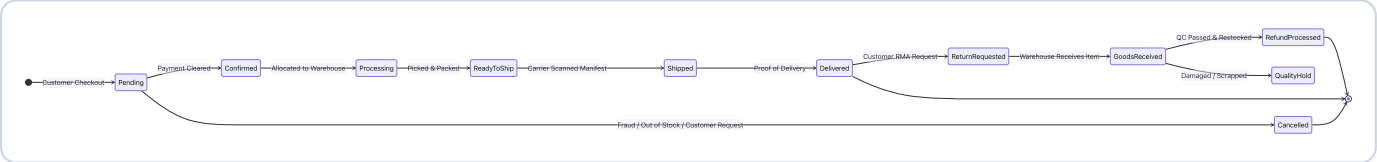
5. Complete Module & Functional Reference

5.1 Executive Dashboard

- **File:** `Main.dc.html`
- **Role:** High-level command center for leadership and operations leads.
- **Core Elements:**
- **Store Switcher:** Live toggle between US, EU, Wholesale, Amazon, or All Stores.
- **KPI Ribbon:** Total Revenue, Net Orders, Average Order Value (AOV), and Conversion Rate with period-over-period delta indicators.
- **Interactive Sales Velocity Chart:** Dual-series daily revenue plot with hover crosshairs, brush zoom, and previous-period comparison dashes.
- **Needs Attention Tasks Queue:** Algorithmic prioritization of actionable alerts (processing bottlenecks, stockout risks, pending refunds, overdue vendor deliveries).
- **Order Pipeline Funnel:** Real-time visibility into order states (Pending -> Confirmed -> Processing -> Ready to Ship -> Shipped) with target cycle times.

5.2 Sales & Order Management

- **Files:** `Orders.dc.html`, `AbandonedCarts.dc.html`, `Returns.dc.html`, `Refunds.dc.html`, `SalesChannels.dc.html`, `DeliveryTracking.dc.html`
- **Role:** End-to-end sales processing, customer support triage, and post-purchase reverse logistics.



- **Functional Highlights:**
- **Faceted Filters & Saved Views:** Filter orders by fulfillment stage, payment status, warehouse source, delivery date, or customer segment. Save custom views with star prioritization.
- **Bulk Action Toolbar:** Select multiple orders for batch fulfillment allocation, tag assignment, PDF picking slip export, or batch cancellation.
- **Abandoned Cart Engine:** Real-time tracking of checkout drop-offs with recovery emails and coupon attribution.
- **Reverse Logistics (RMA):** Integrated returns portal managing reverse shipments, restocking inspection, restocking fees, and instant Stripe/processor refund triggers.

5.3 Catalog & Product Management

- **Files:** `Products.dc.html`, `Categories.dc.html`, `Collections.dc.html`, `Attributes.dc.html`, `ProductReviews.dc.html`
- **Role:** Master product taxonomy, variant matrices, channel merchandising, and customer sentiment.
- **Core Elements:**
- **SKU & Variant Hierarchy:** Manage multi-dimensional products (Size × Color × Material) with individual barcodes (UPC/EAN/GTIN), weights, and cost of goods sold (COGS).
- **Channel Syndication Rules:** Control channel-specific pricing, product titles, and channel listing exclusions (e.g. wholesale-only bundles vs. DTC exclusives).

- **Taxonomy Trees:** Multi-level parent/child categories and rule-based dynamic collections (e.g. "Waterproof Outerwear with Margin > 45%").
- **Review Sentiment Moderation:** Customer review verification, verified buyer badge enforcement, and profanity/link automated moderation filters.

5.4 Inventory & Stock Control

- **Files:** `InventoryOverview.dc.html`, `StockLevels.dc.html`, `LowStock.dc.html`, `StockAdjustments.dc.html`, `StockByWarehouse.dc.html`, `WarehouseTransfers.dc.html`
- **Role:** Multi-echelon stock health monitoring, stock valuation, and rebalancing.

INVENTORY BUCKET	DEFINITION	FORMULA / IMPACT
On Hand	Total physical units physically present in the warehouse bins.	Counted during cycle counts.
Allocated	Units committed to pending/processing customer orders.	Locks units from being sold again.
Available	Units ready to be sold immediately across storefronts.	`On Hand − Allocated`
In Transit	Units shipped from suppliers or moving between warehouses.	Expected arrival dates update velocity forecasts.
Safety Stock / Reorder Point	Threshold trigger that generates automated replenishment requests.	`Lead Time Demand + Buffer Stock`

- **Functional Highlights:**
- **Reorder Automation:** When available stock falls below safety stock, automated low-stock badges trigger supplier purchase orders or warehouse transfers.
- **Audit Cycle Counts:** Full adjustment audit log capturing reason codes (Damaged, Theft, Found, QC Hold) with employee signatures.
- **Transfer Dispatch Board:** Kanban-style transfer management tracking transit status from Draft -> Dispatched -> In Transit -> Received.

5.5 Fulfillment & Logistics

- **Files:** `FulfillmentOverview.dc.html`, `Picking.dc.html`, `Packing.dc.html`, `Shipping.dc.html`, `Warehouses.dc.html`
- **Role:** High-velocity shop-floor warehouse management system (WMS).
- **Core Elements:**
- **Batch Picking Lists:** Algorithmic route-optimized pick lists that group items across multiple orders into serpentine bin paths (Aisle -> Bay -> Shelf -> Bin).
- **Packing Station Console:** Touch-friendly barcode scanning interface verifying SKU matches before cartons are sealed. Box recommendation algorithm calculates dimensional weight.
- **Shipping Slots & Carrier Pickup:** Manifest builder partitioned by carrier (FedEx Ground, UPS 2nd Day, DHL Express, Royal Mail) with live cutoff clocks.

5.6 Procurement & Supplier Operations

- **Files:** `PurchaseOrders.dc.html`, `Suppliers.dc.html`, `Receiving.dc.html`, `GoodsReceived.dc.html`, `PurchaseReturns.dc.html`

- **Role:** Vendor management, inbound purchase orders, dock receiving, and vendor disputes.
- **Core Elements:**
- **PO Lifecycle:** Draft -> Sent to Supplier -> Partially Received -> Received -> Billed.
- **Dock Receiving Console:** Receiving dock operators scan delivered pallets, record carton counts, flag damaged cartons, and generate Goods Received Notes (GRN).
- **Three-Way Matching:** Discrepancy detector matching Purchase Order quantities against Goods Received Notes and Supplier Invoices.
- **Return to Vendor (RTV):** Manages supplier chargebacks for defective goods and return-to-factory freight.

5.7 Customer Relationship Management (CRM)

- **Files:** `Customers.dc.html`, `CustomerGroups.dc.html`, `Segments.dc.html`, `CustomerActivity.dc.html`
- **Role:** 360-degree customer intelligence, lifetime value (LTV) analytics, and automated cohort segmentation.
- **Core Elements:**
- **Unified Customer Ledger:** Tracks order history, total spend, address book, payment tokens, tax exemption certificates, and customer support tickets.
- **RFM Segmentation Engine:** Segments customers into Champions, Loyalists, Potential Loyalists, At Risk, and Dormant based on Recency, Frequency, and Monetary parameters.
- **B2B Groups & Tiers:** Assign custom pricing tiers (Wholesale Tier 1, VIP Creators, Retail) and payment terms (Net 30, Pre-paid credit card).

5.8 Marketing & Promotional Engine

- **Files:** `Campaigns.dc.html`, `Discounts.dc.html`, `Coupons.dc.html`, `Promotions.dc.html`
- **Role:** Revenue acceleration, campaign tracking, and discount validation.
- **Core Elements:**
- **Automated Cart Rules:** Free shipping on orders over \$75, Buy-One-Get-One (BOGO), and tiered quantity discounts.
- **Coupon Engine:** Single-use vs. multi-use promotional codes, creator affiliate tracking, and maximum usage quotas.
- **Marketing Campaign Attribution:** Revenue and ROI calculation per channel (Email, SMS, Meta Ads, Google Ads).

5.9 Finance, Accounting & Ledger

- **Files:** `FinanceOverview.dc.html`, `FinancialReports.dc.html`, `Payments.dc.html`, `Expenses.dc.html`, `Transactions.dc.html`
- **Role:** Financial compliance, P&L reporting, merchant reconciliation, and cash flow forecasting.
- **Core Elements:**
- **Automated P&L Statements:** Gross Sales -> Discounts -> Net Sales -> COGS -> Gross Margin -> Operating Expenses (Opex) -> EBITDA.
- **Payment Processor Settlement:** Reconciles gross sales against Stripe/PayPal transaction fees and net bank payouts.
- **Accounts Receivable (A/R):** Manages wholesale Net 30/45 invoice schedules with aging buckets (Current, 1-30 days overdue, 31-60 days overdue).
- **Bank Ledger Reconciliation:** One-click matching of bank feed transactions against invoices, payouts, and vendor expense claims.

5.10 Business Intelligence & Analytics

- **Files:** `SalesAnalytics.dc.html`, `ProductAnalytics.dc.html`, `CustomerAnalytics.dc.html`, `InventoryAnalytics.dc.html`
- **Role:** Strategic intelligence powered by `viz.js` components (`VizLine`, `VizColumns`, `VizRank`, `VizHeat`, `VizScatter`).
- **Core Elements:**
- **Product Velocity vs. Margin Scatter:** Quadrant analysis identifying Star products (high volume, high margin) vs. Cash Cows, Question Marks, and Dogs.
- **Cohort Retention Heatmaps:** Visualizes month-over-month customer repurchase rates.
- **Inventory Aging & Turn Analysis:** Identifies dead stock (>90 days in warehouse) and calculates days of inventory remaining (DIO).

5.11 Administration, Security & Governance

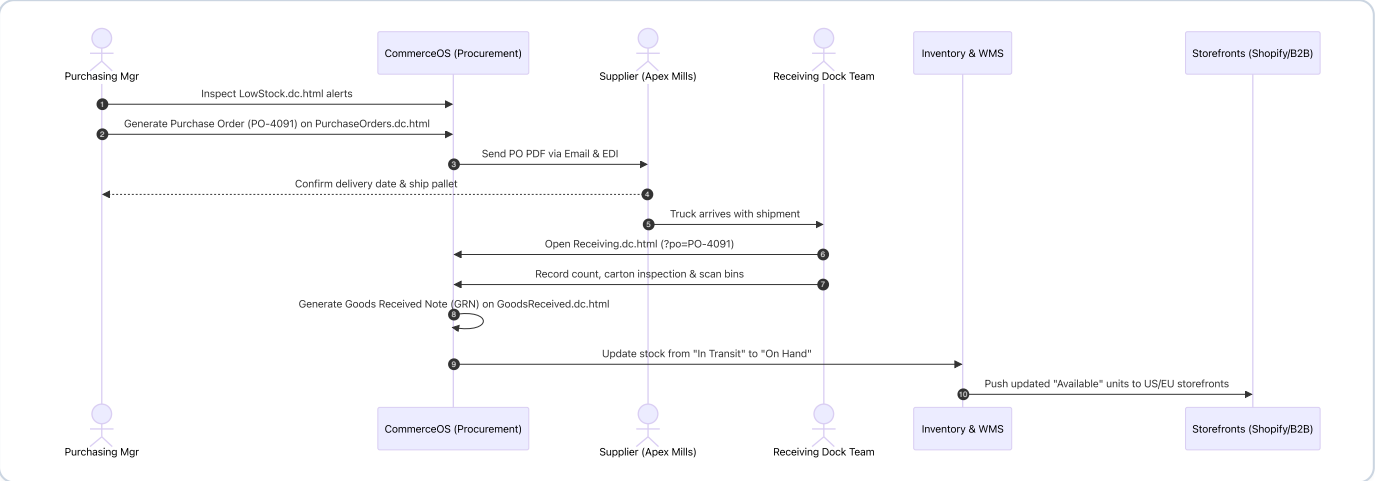
- **Files:** `Users.dc.html`, `Integrations.dc.html`, `Settings.dc.html`, `Profile.dc.html`, `Preferences.dc.html`, `Help.dc.html`, `SignIn.dc.html`
- **Role:** Enterprise security, granular permissions, integration ecosystems, and user preferences.
- **Core Elements:**
- **Granular RBAC:** Permissions matrix partitioned across Read, Edit, Create, and Admin access per module.
- **Integrations Gateway:** Webhooks, REST API keys, and pre-built connectors (Shopify, Amazon, Stripe, FedEx, Slack).
- **User Ergonomics & Accessibility:** Preferences for dark mode, reduced motion, dense table layout, and keyboard shortcuts.

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6. End-to-End Business User Flows

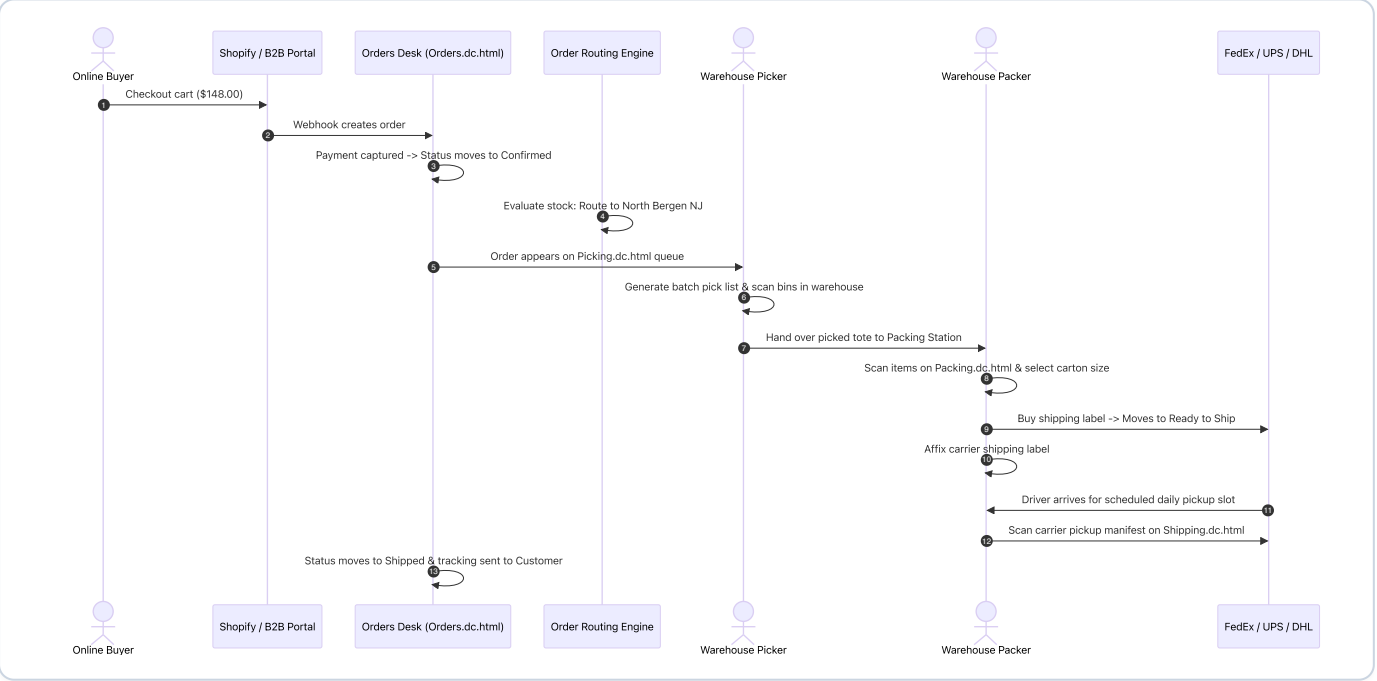
Flow 1: Procurement, Goods Receipt & Stock Availability

When stock drops below threshold, this flow manages restocking from supplier purchase order to active storefront availability.



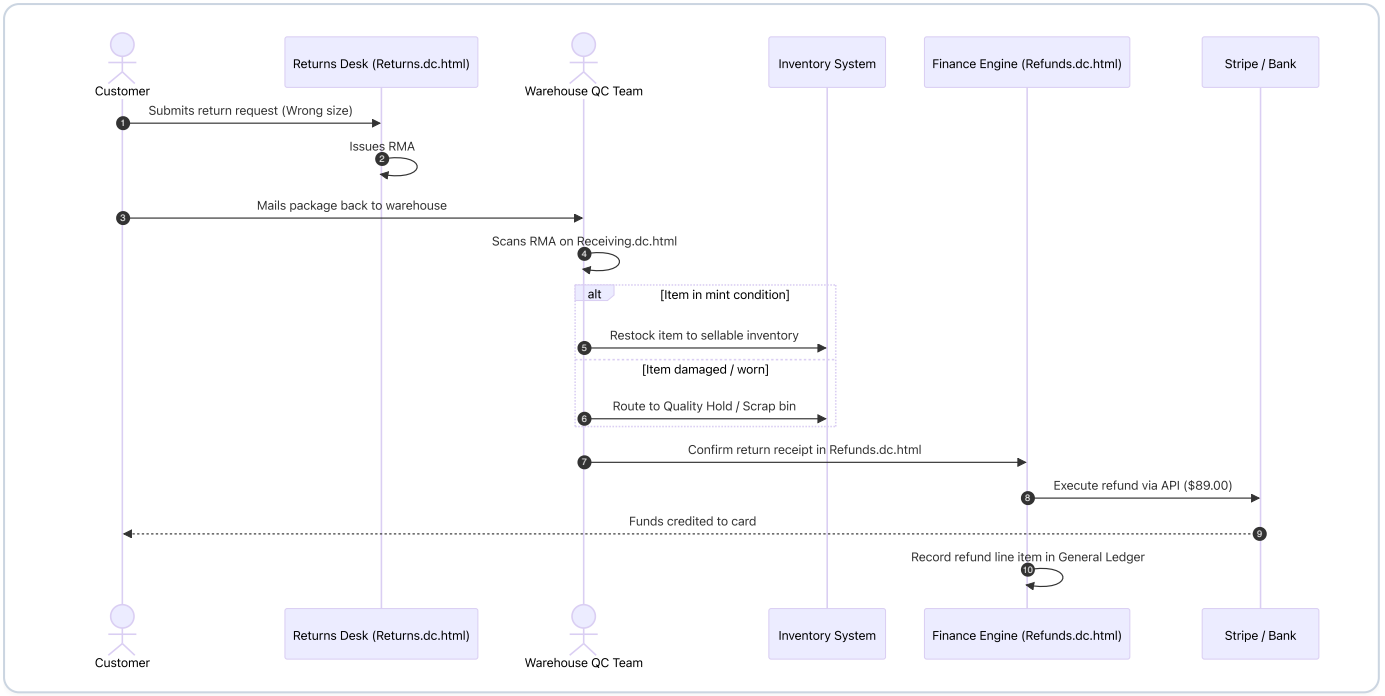
Flow 2: Omnichannel Order-to-Cash & Fulfillment

The critical path from customer purchase to carrier parcel handover.



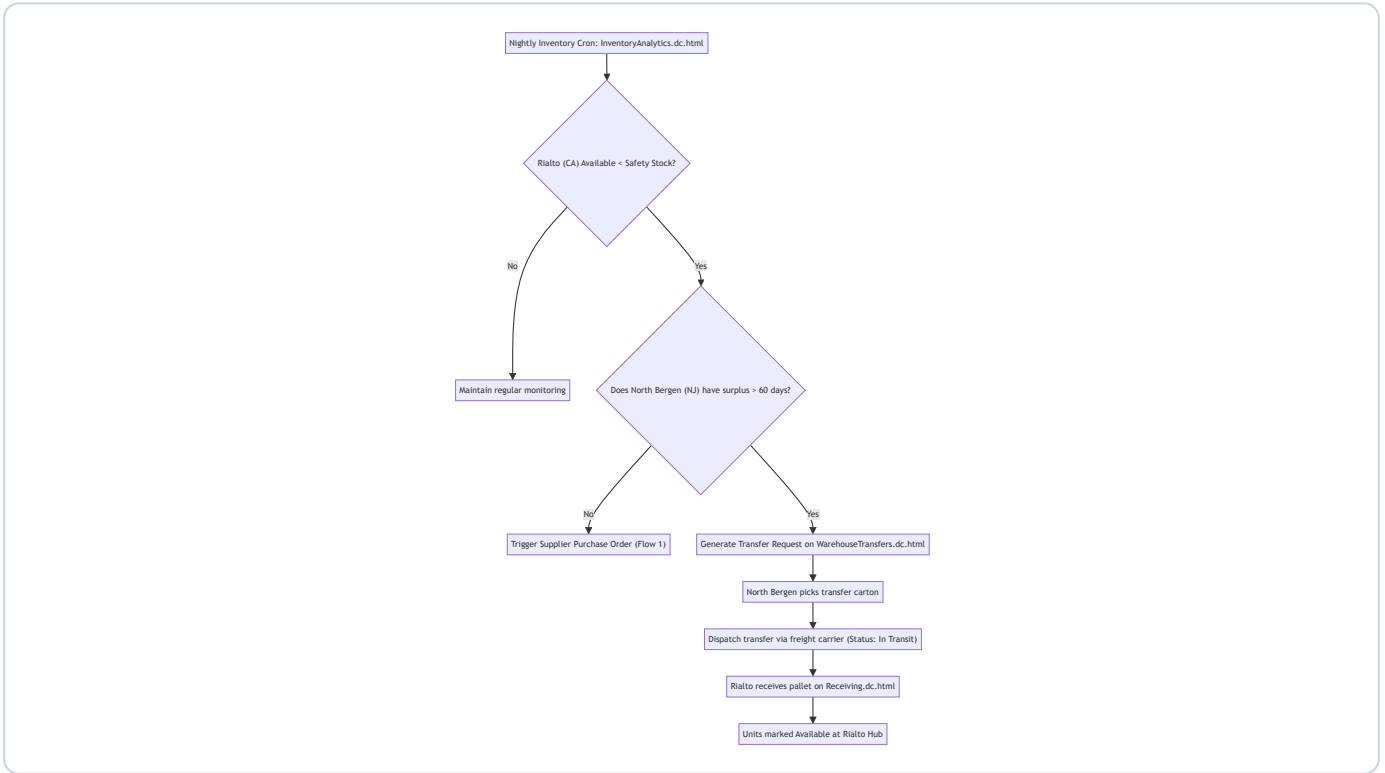
Flow 3: Post-Purchase RMA Return, Inspection & Refund

Customer return, return merchandise authorization (RMA), quality inspection, and refund settlement.



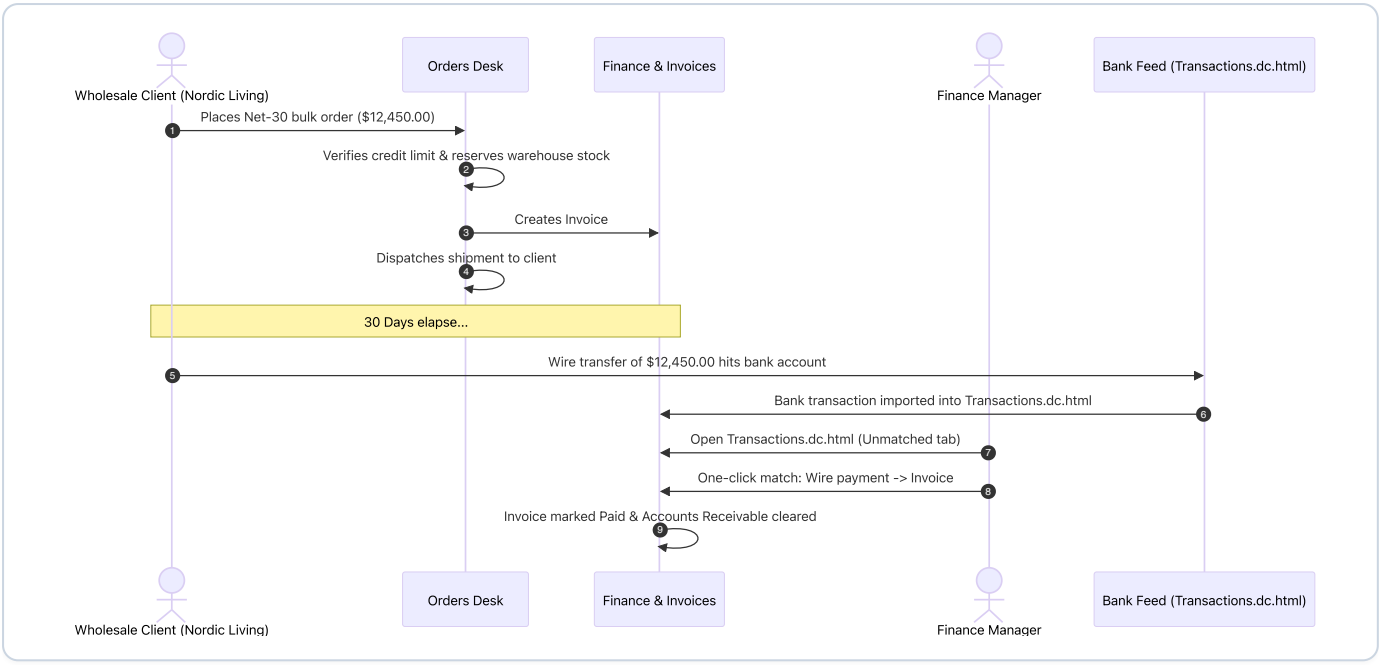
Flow 4: Low-Stock Detection & Inter-Warehouse Transfer

Automated balance of inventory between primary hubs and regional nodes.



Flow 5: B2B Wholesale Invoicing & Ledger Reconciliation

Wholesale client order, Net-30 invoicing, and bank statement matching.


Wholesale Client (Nordic Living)
Orders Desk
Finance & Invoices
Finance Manager
Bank Feed (Transactions.dc.html)
1 Places Net-30 bulk order ($12,450.00)
2 Verifies credit limit & reserves warehouse stock
3 Creates Invoice
4 Dispatches shipment to client
30 Days elapse...
5 Wire transfer of $12,450.00 hits bank account
6 Bank transaction imported into Transactions.dc.html
7 Open Transactions.dc.html (Unmatched tab)
8 One-click match: Wire payment -> Invoice
9 Invoice marked Paid & Accounts Receivable cleared
Wholesale Client (Nordic Living)
Orders Desk
Finance & Invoices
Finance Manager
Bank Feed (Transactions.dc.html)

7. Security, Governance & Audit Trails

Granular Role-Based Access Control (RBAC) Matrix

Permissions are enforced at the module and action level via `management.js`:

MODULE	OPERATIONS LEAD	WAREHOUSE PICKER	FINANCIAL CONTROLLER	CUSTOMER SUPPORT	SYSTEM ADMIN
Dashboard	Full View	Restricted	Full View	View Only	Full Admin
Sales & Orders	Full Edit	View Pick List	View Only	Edit Notes & RMA	Full Admin
Inventory & Stock	Full Edit	Read / Scan	View Valuation	View Availability	Full Admin
Fulfillment / WMS	Full Edit	Pick & Pack Scan	View Only	View Tracking	Full Admin
Procurement	Full Edit	Goods Receipt	View PO Invoices	No Access	Full Admin
CRM & Customers	Full Edit	No Access	View Billing	View & Edit Profile	Full Admin
Marketing Engine	Full Edit	No Access	View Spend	View Promo Codes	Full Admin
Finance & Ledger	View Only	No Access	Full Accounting	View Payment Status	Full Admin
Analytics & BI	Full View	No Access	Full View	View Order KPIs	Full Admin
Settings & Users	View Profile	View Profile	Billing Settings	View Profile	Full Admin

Immutable Audit Log

All mutations (stock manual overrides, cancellations, role privilege escalations, and refund authorizations) are recorded in `Users.dc.html?tab=log` capturing:

- Timestamp (UTC ISO 8601)
- User ID & Display Name
- IP Address & User Agent
- Action Verb & Target Entity ID
- Delta Payload (`before` vs. `after` attributes)

8. Keyboard Shortcuts & Power-User Reference

CommerceOS features dedicated keyboard navigability for high-volume desktop workflows:

SHORTCUT	CONTEXT	FUNCTION
`⌘` + `K` / `Ctrl` + `K`	Global	Open Command Palette search modal across orders, SKUs, customers, and routes
`[`	Global	Toggle sidebar persistence (Expand 240px ↔ Collapse 64px Icon Rail)
`?`	Global	Open Help & Shortcut cheat sheet modal
`Esc`	Modal / Panel	Dismiss active dialog, drawer panel, or search palette
`Enter`	Command Palette	Navigate directly to the highlighted record
`J` / `K`	Command Palette	Navigate search items up/down
`Alt` + `T`	Global	Toggle Light Mode / Dark Mode theme

TIP

Developer & Customization Reference:
To extend CommerceOS with custom pages, link `ds/commerceos/tokens.css`, `bundle.css`, `shell.css`, and `list.css`. Wrap page content with `<div class="app cx-shell">` and enforce the 20px card rhythm (`--space-20`).